# Sarah Thomas

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## Education

**Vellore Institute of Technology** September 2023 – May 2027
*B.Tech in Computer Science – Specialization in Cloud Computing and Automation* *CGPA: 8.81/10*

**St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh** March 2022 – May 2023
*All India Senior School Certificate Examination* *CGPA: 7.88/10*

**St. Joseph's Convent Sr. Sec. School, Sagar, Madhya Pradesh** March 2020 – June 2021
*All India Secondary School Examination* *CGPA: 9.2/10*

## Experience

**MPOnline Limited (Govt. of Madhya Pradesh & TCS Joint Venture)** 20 May 2026 – 31 Aug 2026
*Software Engineering & AI Foundation Intern* *Remote*

- Engineered robust database architectures and applied advanced object-oriented programming paradigms, executing rigorous unit/integration testing under Agile workflows.
- Collaborated in a cross-functional 7-member team to architect an AI-Powered Smart Logistics System using ASP.NET Core, integrating EF Core, JWT role-based access, and SignalR for real-time tracking.
- Implemented generative AI frameworks, Retrieval-Augmented Generation (RAG) models, and automated backend workflow integrations using n8n.

## Technical Skills

**Cloud & DevOps**: AWS (Lambda, S3, API Gateway, EC2, IAM, CloudWatch), Docker, Docker Compose, Prometheus, Terraform, GitHub Actions, Git, GitHub, Boto3

**Languages & Frameworks**: Java, Python, Linux/Bash, FastAPI, REST APIs, AIOps, Scikit-Learn, OpenCV, ASP.NET Core, C#

**Databases & Tools**: MongoDB, Microsoft SQL Server, OpenAI API, Pandas, Matplotlib, EF Core, SignalR, n8n

## Projects

**AI Incident & Log Analysis Assistant**
*Python, FastAPI, OpenAI API* *github.com/THOMAS-SARAH/exception_analyzer*

- Architected a RESTful AI incident response microservice utilizing FastAPI and OpenAI to process server log traces and automate Root-Cause Analysis (RCA) report generation.
- Formulated structured prompt engineering workflows to analyze system telemetry errors, delivering actionable remediation steps for technical operations teams.

**Hybrid Ensemble Model for Enhanced Cancer Cell Classification**
*Python, Scikit-Learn, OpenCV, Pandas, Matplotlib* *github.com/THOMAS-SARAH/CANCER_CELL_DETECTION*

- Developed a hybrid ensemble learning model combining multiple classifiers to distinguish cancerous cells, optimizing data preprocessing and extraction pipelines via OpenCV.
- Evaluated complex model performance using accuracy and AUC metrics, visualizing comparative validation results with Pandas and Matplotlib.

**AI-Driven Cloud Resource & Cost Optimizer**
*Prometheus, Node Exporter, Terraform, AWS, Linux/Bash, FastAPI* *github.com/THOMAS-SARAH/multi-region-aws-telemetry-pipeline*

- Deployed a real-time infrastructure telemetry pipeline using Prometheus and Node Exporter to monitor system performance, CPU/memory utilization, and cloud workload health.
- Provisioned multi-region cloud infrastructure via modular Terraform templates, integrating FastAPI backend endpoints for automated resource optimization and anomaly detection.

## Certifications and Extra-Curricular

**Certifications:** ServiceNow University Virtual Internship Program; NPTEL – Internet Of Things, Cloud Computing and Distributed Systems; AWS Educate – Introduction to Generative AI.
**Extra-Curriculars:** Education Minister in School Council, Best Girl Student of my Batch 2022-23, Club Coordinator of E-Green Club.